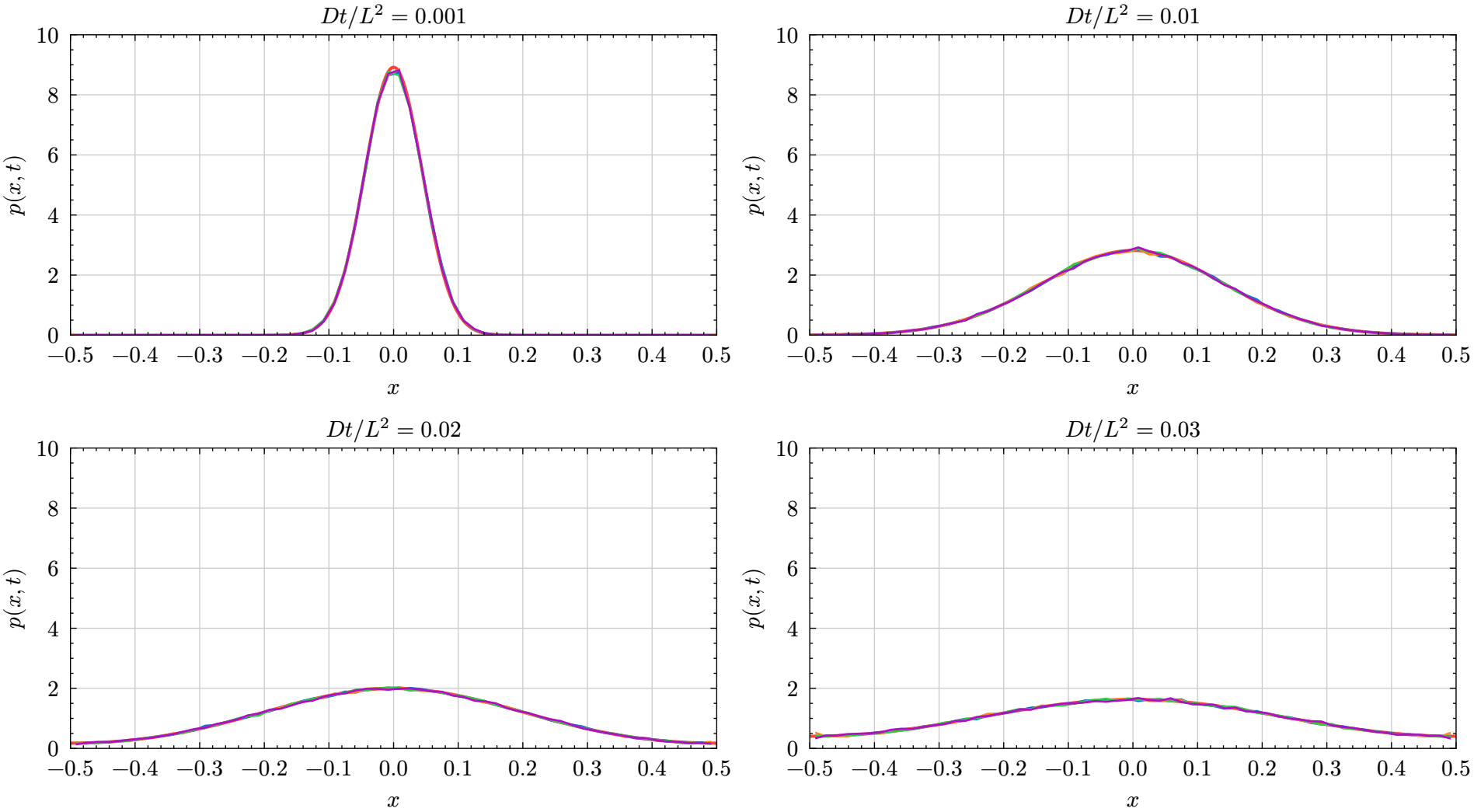
Kronberger P1_4b: Comparing wall-handling methods


$Dt/L^2 = 0.001$
$Dt/L^2 = 0.01$
$Dt/L^2 = 0.02$
$Dt/L^2 = 0.03$
$p(x,t)$
x
0
2
4
6
8
10
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5

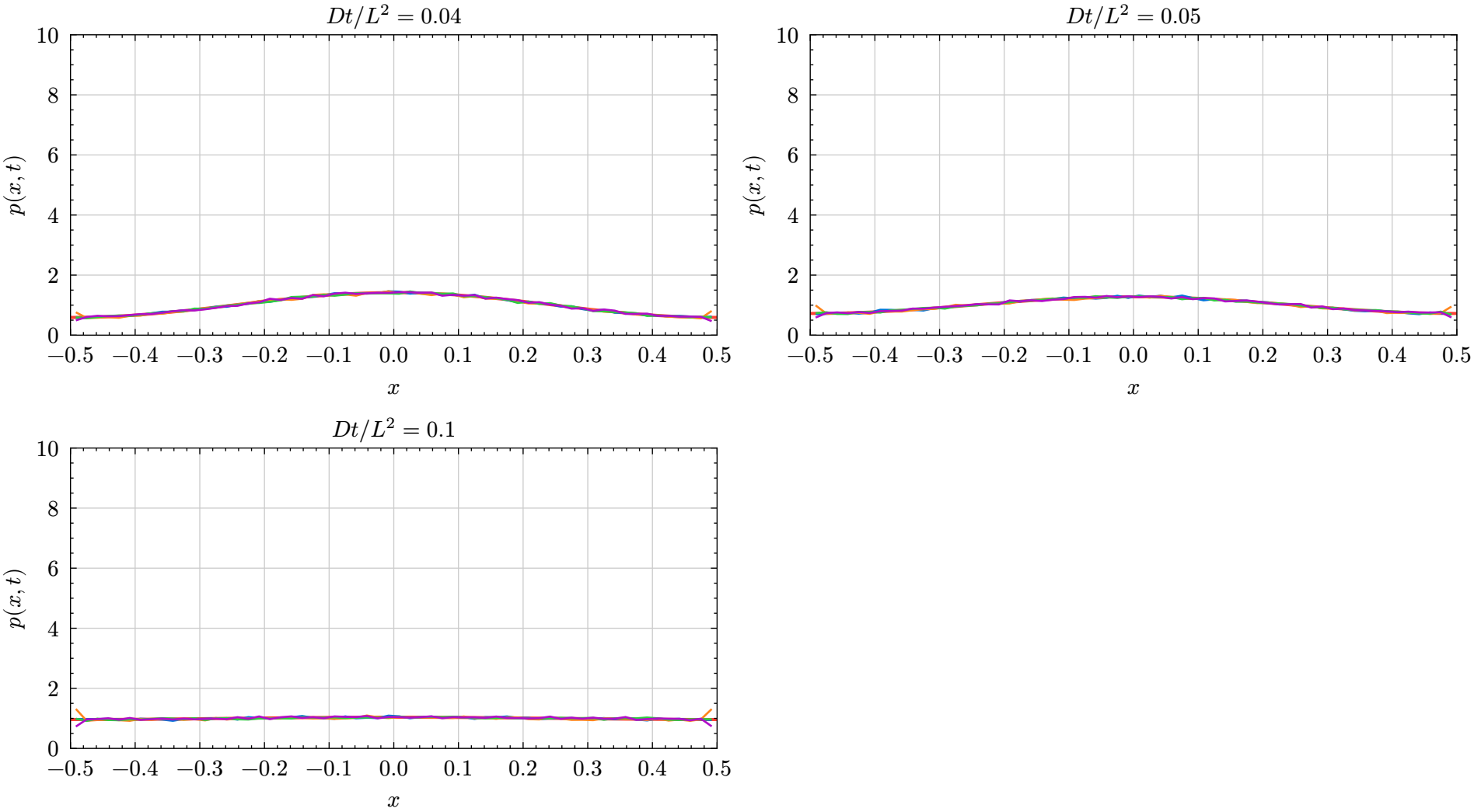


Figure 1: Comparison of four wall-handling methods. “Reflect” correctly reproduces the analytical solution. “Stop at wall” accumulates density at the boundaries. “Don’t move” also leads to boundary accumulation. “Redraw” biases the distribution toward the center by rejecting large steps near walls.